Lecture 35 — Eigenvectors as persistent directions

Mon Nov 16

Last updated: October 8, 2026.

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Beginning of Strang §6.1 — $Ax = \lambda x$, eigenvectors, and dynamics. Related objectives: Lectures. Schedule.

Bring paper. Warm-up while attendance is taken, then fill-in notes, board examples, and **Your turn**.

Callback: Lab 4 asked why $B = \text{diag}(1.2, 0.5)$ grows from some starts and decays from others. Today we name the missing idea: eigenvectors.

Learning objectives

By the end of class, students should be able to:

1. Explain geometrically what makes v an eigenvector.
2. Verify a proposed eigenpair.
3. Find λ when an eigenvector is given.
4. Connect eigenvectors to repeated transformations and steady directions.
5. Recognize that a rotation may have no real eigenvector, motivating complex numbers.

Warm-up

Lab 4 mystery matrix:

$$B = \begin{bmatrix} 1.2 & 0 \\ 0 & 0.5 \end{bmatrix}.$$

(a) Compute $B \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $B \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Is each output a scalar multiple of the input?

(b) Starting from $x_0 = (1, 0)$ vs $y_0 = (0, 1)$, which trajectory grows and which decays? Why, in one sentence using part (a)?

Fill in

Eigenpair. A nonzero vector v is an **eigenvector** of square A with **eigenvalue** λ if

$$Av = _____.$$

Geometrically: A sends v to a vector on the _____ line through v (scaled by λ; if $\lambda < 0$, also flipped).

Verify. Given a candidate pair (λ, v), compute Av and λv; they must _____.

Find λ from v. If $v \neq 0$ is known to be an eigenvector and some component $v_i \neq 0$, then

$$\lambda = \frac{(Av)_i}{v_i}$$

(same ratio for every nonzero component).

Dynamics. If $x_0 = v$ is an eigenvector with eigenvalue λ, then

$$x_t = A^t v = _____.$$

The direction of v is **persistent**: each step only rescales (and maybe flips). That is why Lab 4's two axis starts behaved so differently.

No real eigenvector. A pure rotation (e.g. 90°) moves every real nonzero vector off its own line. Then there is _____ real eigenpair — we will need complex numbers on Wednesday.

Worked in class

W1. Geometry. Sketch $v = (1, 1)$ and Av for $A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$. Is v an eigenvector? What about $e_1 = (1, 0)$?

W2. Verify. Check whether $(\lambda, v) = (3, (1, 1))$ is an eigenpair of

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Compute both Av and λv.

W3. Find λ. You are told that $v = (2, -1)$ is an eigenvector of

$$A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}.$$

Find λ. Then confirm $Av = \lambda v$.

W4. Lab 4 explained. For $B = \text{diag}(1.2,\ 0.5)$:

(a) Name two real eigenpairs (use the standard basis).

(b) If $x_0 = c_1 e_1 + c_2 e_2$, what is $x_t = B^t x_0$? Which term dominates for large t when $c_1 \neq 0$?

W5. Rotation. Let $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ (90° CCW). For a generic $v = (a, b) \neq 0$, is Rv a real multiple of v? What does that say about real eigenvectors of R?

Your turn

Y1. Verify or reject: is $(\lambda, v) = (-1, (1, -1))$ an eigenpair of $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$?

Y2. Given that $v = (1, 2)$ is an eigenvector of $A = \begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix}$, find λ.

Y3. True or false, with a one-line reason: if $Av = 2v$ and $v \neq 0$, then $A^3 v = 8v$.

Y4. In one sentence: why did Lab 4's diagonal mystery need eigenvectors (not just "look at the plot")?

Answers (Your turn)

Y1. $Av = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ and $(-1)v = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$. Yes — eigenpair.

Y2. $Av = \begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. Ratio from the first component: $\lambda = 3/1 = 3$. Check second: $4 = 3 \cdot 2$. So $\lambda = 3$.

Y3. **True.** $A^3v = A^2(Av) = A^2(2v) = A(4v) = 8v$ (or $A^tv = \lambda^t v$).

Y4. Sample: The plot shows growth vs decay by start direction; eigenvectors name the persistent directions (and their growth factors λ) that explain why.